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[5352]-178

S.E. (Information Technology) (II Sem.) EXAMINATION, 2018

DATA STRUCTURES AND FILES

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Answer *four* questions in all.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Assume suitable data, if necessary.

1. a) Explain stack as an ADT. List applications of stack. 6

b) Write pseudo 'C++' code to implement linear queue using array. 6

OR

2. a) Clearly indicate the contents of stack during conversion of given infix expression into postfix expression : $A \wedge B * C - D + E / F / (G + H)$.

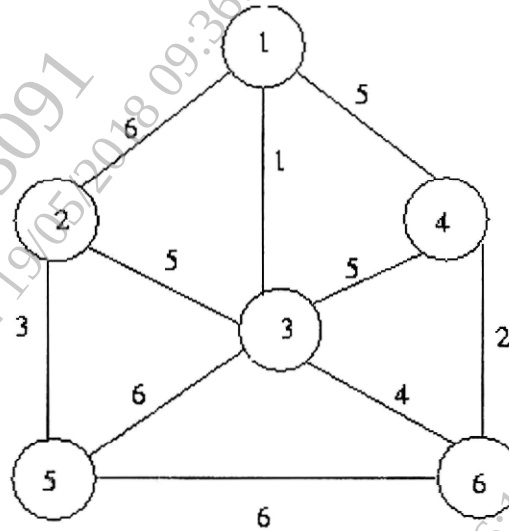
Consider $\wedge$ as exponent operator. 6

b) What is circular queue ? How is it better than linear queue ? Explain priority queue in brief. 6

3. a) What is Threaded Binary Tree ? Explain inorder threaded binary tree with example. 6

P.T.O.

- b) For the given graph below, draw the minimum spanning tree step wise using Krushkal's method. 6



OR

4. a) Define Binary Search Tree. Draw the BST stepwise for given nodes by assuming node's incoming sequence from left to right. 6
i, h, b, m, e, c, f, a, d, k
- b) What is topological sorting? Write an algorithm for topological sorting with its time complexity. 6
5. a) What is hashing? Explain various hash collision resolution techniques. 6
b) Obtain an AVL tree by inserting one data element at a time in the following sequence. Label the rotations appropriately at each stage. 8
3,2,1,4,5,6,7, 16,15
- OR
6. a) Sort the following numbers in ascending order using heap sort. Show the sorting stepwise. 6
5, 3, 17, 10, 84, 19, 6, 22, 9
- b) Why is Huffman code used for data compression? Draw a Huffman tree for the given sentence & find the corresponding Huffman codes for every characters present in sentence. (Note: Count "spaces" & "." Operator also as characters) 8
Eerie eyes seen near lake.

7. a) What is file? Compare features of different types of file organization. 6
- b) Write C++ program to perform the following operations on sequential file. 6
- i. Create records
 - ii. Delete record with given key value
 - iii. Display records
- OR
8. a) Explain fread(), fwrite(), seekg() functions with example. 6
- b) Write C++ program to perform the following operations on index sequential file. 6
- i. Create records.
 - ii. Display records.